Practice Test: 5.1–5.3 (Composition, Inverses, Exponentials)

Name: _____ Date: _____

About 50 minutes. Calculator only where marked.

1. (5.1) $f(x) = x^2 - 1$ and $g(x) = 3x + 2$. Find:

- (a) $f(g(1))$ (b) $g(f(1))$ (c) $f(f(2))$
- (d) $f(g(x))$ (e) $g(f(x))$

2. (5.1) Use the table.

x	1	2	3	4
$f(x)$	3	4	1	2
$g(x)$	2	1	4	3

Find $f(g(1))$, $g(f(1))$, $f(f(3))$ and $g(g(4))$.

3. (5.1) Find each composite and its domain.

- (a) $f(x) = \sqrt{x}$, $g(x) = x - 5$: find $f(g(x))$ and $g(f(x))$, each with its domain.
- (b) ★ $f(x) = \dfrac{2}{x-1}$, $g(x) = \dfrac{4}{x}$: find $f(g(x))$ and its domain.

4. (5.1) Find f and g so that $h(x) = f(g(x))$. (a) $h(x) = (2x+5)^3$ (b) $h(x) = \sqrt{x^2+1}$ (c) $h(x) = \dfrac{1}{x-7}$

5. (5.1) A company makes $n(t) = 25t$ shirts in t hours, and it costs $C(n) = 10n + 50$ dollars to make n shirts.

- (a) Find $C(n(t))$ and say what it represents, with units.
- (b) Find $C(n(4))$ and interpret it.

6. (5.2) One-to-one or not? Explain.

- (a) $\{(1,4),(2,5),(3,4)\}$ (b) $\{(-1,2),(0,3),(2,7)\}$
- (c) $y = x^2$ (d) $y = x^3$ (e) $y = |x|$ (f) $y = 2^x$

7. (5.2) f passes through $(-2, 1)$, $(0, 3)$ and $(4, 7)$, has domain $[-2, 4]$ and range $[1, 7]$.

- (a) List three points on f^{-1}.
- (b) If $f(5) = 11$, what is $f^{-1}(11)$?
- (c) What are the domain and range of f^{-1}?

8. (5.2) Use the table: find $f^{-1}(11)$, $f^{-1}(5)$ and $f^{-1}(f(3))$.

x	0	1	2	3
$f(x)$	5	8	11	14

9. (5.2) Find the inverse.

- (a) $f(x) = 5x - 3$
- (b) $f(x) = x^3 + 4$
- (c) ★ $f(x) = \dfrac{x+1}{x-2}$. Also give the domain and range of f.

10. (5.2) Show that $f(x) = 2x - 6$ and $g(x) = \frac{x}{2} + 3$ are inverses.

11. (5.2) ★ $f(x) = (x-2)^2$ is not one-to-one.

- (a) Restrict its domain so it is, keeping the right half.
- (b) Find f^{-1} and give its domain and range.

12. (5.2) The graph of f contains the points $(0, 1)$, $(1, 2)$ and $(2, 4)$. Sketch f and f^{-1} on the same axes along with the line $y = x$. Where do the points go?

13. (5.3) $f(x) = 3 \cdot 2^x$. Find $f(0)$, $f(3)$ and $f(-1)$.

14. (5.3) For each, give the initial value, growth or decay, and the percent rate. (a) $500(1.03)^t$ (b) $80(0.6)^t$

15. (5.3) $f(x) = 2^{x-1} + 3$. Describe the transformation from 2^x, and give the domain, range, horizontal asymptote and y-intercept.

16. (5.3) Find the exponential function $y = ab^x$ through each pair of points. (a) $(0, 4)$ and $(2, 36)$ (b) $(1, 10)$ and $(3, 2.5)$

17. (5.3) Solve. (a) $2^{x+1} = 16$ (b) $9^x = 27$ (c) $\left(\frac{1}{2}\right)^x = 8$ (d) ★ $3^{x^2} = 81$

18. (5.3) \$5,000 is invested at 6% for 5 years. Find the balance compounded quarterly and compounded continuously. (Calculator)

19. (5.3) A 200 g sample has a half-life of 10 years. Write a model and find how much is left after 25 years. (Calculator)

20. (5.1-5.3, putting it together) $f(x) = 2^x$ and $g(x) = x - 3$.

- (a) Find $h(x) = f(g(x))$. Describe it as a transformation of 2^x.
- (b) Is h one-to-one? How do you know?
- (c) $h(5) = 4$. What point must be on the graph of h^{-1}?
- (d) What is the horizontal asymptote of h? What vertical line would the graph of h^{-1} approach instead?

Answer Key

1. (a) $g(1) = 5$, so $f(5) = 24$. (b) $f(1) = 0$, so $g(0) = 2$. (c) $f(2) = 3$, so $f(3) = 8$. (d) $9x^2 + 12x + 3$. (e) $3x^2 - 1$.
2. $f(g(1)) = f(2) = 4$. $g(f(1)) = g(3) = 4$. $f(f(3)) = f(1) = 3$. $g(g(4)) = g(3) = 4$.
3. (a) $f(g(x)) = \sqrt{x-5}$, domain $x \geq 5$. $g(f(x)) = \sqrt{x} - 5$, domain $x \geq 0$. (b) $f(g(x)) = \dfrac{2}{\frac{4}{x} - 1} = \dfrac{2x}{4-x}$. Domain: $x \neq 0$ (for g) and $\frac{4}{x} \neq 1$, so $x \neq 4$. All reals except 0 and 4.
4. (a) $g = 2x + 5$, $f = x^3$. (b) $g = x^2 + 1$, $f = \sqrt{x}$. (c) $g = x - 7$, $f = \frac{1}{x}$. (Other splits can work too.)
5. (a) $C(n(t)) = 10(25t) + 50 = 250t + 50$: the cost in dollars of running production for t hours. (b) $C(n(4)) = 1050$: 4 hours of production (100 shirts) costs \$1,050.
6. (a) No: the output 4 comes from both 1 and 3. (b) Yes: no repeated outputs. (c) No: fails the horizontal line test. (d) Yes: always increasing. (e) No: $|2| = |-2|$. (f) Yes: always increasing.
7. (a) $(1, -2)$, $(3, 0)$, $(7, 4)$. (b) 5. (c) Domain $[1, 7]$, range $[-2, 4]$.
8. $f^{-1}(11) = 2$, $f^{-1}(5) = 0$, $f^{-1}(f(3)) = 3$
9. (a) $f^{-1}(x) = \dfrac{x+3}{5}$. (b) $f^{-1}(x) = \sqrt[3]{x-4}$. (c) $f^{-1}(x) = \dfrac{2x+1}{x-1}$. Domain of f: $x \neq 2$. Range of f: $y \neq 1$ (the domain of f^{-1}).
10. $f(g(x)) = 2\left(\frac{x}{2} + 3\right) - 6 = x$ and $g(f(x)) = \frac{2x-6}{2} + 3 = x$. Both give x, so they're inverses.
11. (a) $x \geq 2$. (b) $f^{-1}(x) = 2 + \sqrt{x}$, domain $x \geq 0$, range $y \geq 2$.
12. f^{-1} contains $(1, 0)$, $(2, 1)$ and $(4, 2)$. Each point flips across $y = x$, at the same distance from the line on the other side.
13. $f(0) = 3$, $f(3) = 24$, $f(-1) = 1.5$
14. (a) Initial 500, growth, 3%. (b) Initial 80, decay, 40%.
15. Right 1, up 3. Domain all reals, range $y > 3$, horizontal asymptote $y = 3$, y-intercept $2^{-1} + 3 = 3.5$.
16. (a) $a = 4$, $b^2 = 9$, so $b = 3$: $y = 4 \cdot 3^x$. (b) $y = 20(0.5)^x$.
17. (a) $2^{x+1} = 2^4$, so $x = 3$. (b) $3^{2x} = 3^3$, so $x = \frac{3}{2}$. (c) $2^{-x} = 2^3$, so $x = -3$. (d) $x^2 = 4$, so $x = \pm 2$.
18. Quarterly: $5000(1.015)^{20} \approx \$6{,}734.28$. Continuous: $5000e^{0.3} \approx \$6{,}749.29$.
19. $A = 200\left(\frac{1}{2}\right)^{t/10}$, and $A(25) = 200(0.5)^{2.5} \approx 35.4$ g
20. (a) $h(x) = 2^{x-3}$: 2^x shifted right 3. (b) Yes: it's always increasing, so it passes the horizontal line test. (c) $(4, 5)$. (d) h has horizontal asymptote $y = 0$. Reflecting over $y = x$ turns it into the vertical asymptote $x = 0$ for h^{-1}.